Figure 14.11 DFS in action. The algorithm *implicitly* (usually, in recursive implementations – otherwise explicitly) maintains a stack with the next vertices to be traversed. In case the stack is kept explicitly, it is also necessary to remember which edges has been travelled.

To complete the traversal of the graph, we need to start another traversal in one of the remaining vertices, the ones that haven't been visited before: this is shown in figure 14.13 where the new traversal starts from vertex v_3, and avoids the 3 vertices that hadn't already been visited in the first run.

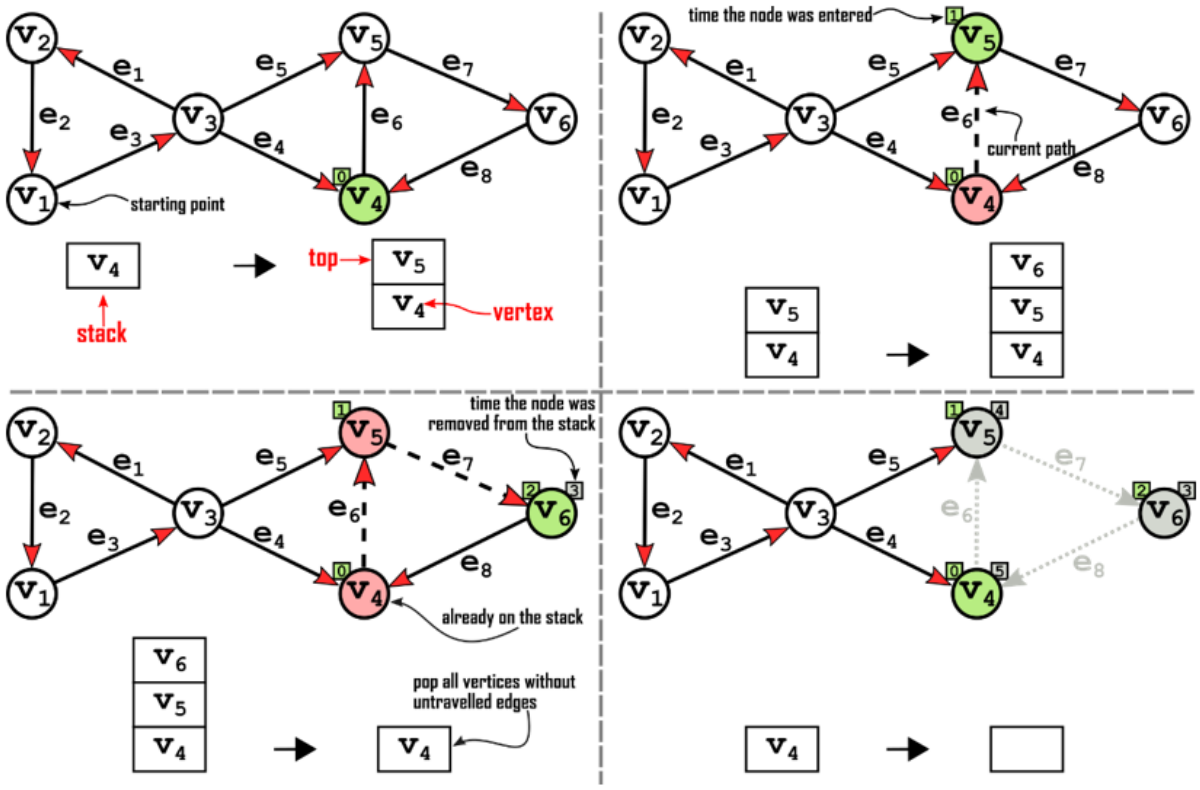


Figure 14.12 DFS in action: the starting point is crucial in graph's traversal, both DFS and BFS